Venkata Sai Surya Rao Beeraka

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Education

University of Illinois at Chicago May 2025
Master of Science in Computer Science *Chicago, USA*
Coursework: DSA, Distributed Systems, Database Systems, Object-Oriented Design, Human Computer Interaction, AI

Jain University May 2021
Bachelor of Technology in Computer Science and Engineering *Bangalore, India*

Experience

University of Illinois at Chicago | Full-Stack Software Engineer Jan 2025 - Present

- Rebuilt UIC's legacy HR form system for **14,000+ students** in five departments, migrating from PHP/larvel to a Spring Boot backend and React/Redux front end with SharePoint auto-requisitions, cutting onboarding time **40%**.
- Designed **reusable** and **configurable UI** components using React.js, TypeScript, and Tailwind CSS. Enabled dynamic form creation with custom field types, reducing development overhead and defect rates by **55%**.
- Redesigned database schema with detailed **ER modeling** and optimized PostgreSQL performance using GIN and composite indexes, accelerating report generation and REST API response times from **3s to 800ms**.

Carelon Global Solutions, India | Software Engineer Oct 2022 - Jul 2023

- Refactored a 250 K-line monolith into **12 Dockerized Spring Boot microservices** orchestrated with Kafka, cutting release lead-time **80%** and **tripling** ingestion speed for 15,000+ monthly XML healthcare feeds.
- **Collaborated** with QA and peer engineers to embed 200+ JUnit tests and static checks into Jenkins CI/CD pipelines, instituting formal code-review gates that **boosted test reliability 20%** and eliminated build failures caused by misformatted reports.
- Authored and versioned **RESTful APIs**, then deployed them on **AWS EC2 + RDS** read-replicas introduced retry logic and Hystrix-style circuit breakers that kept **p95 latency under 150ms during 5x** traffic surges.
- Reengineered SQL Server ETL pipelines (T-SQL, SSIS) with table partitioning, covering indexes, and batched writes, shrinking runtime **85% (3 hours → 45 minutes)** and **saving 12 CPU-hours/day** across 10,000+ claim transactions.

Carelon Global Solutions, India | Associate Software Engineer Aug 2021 - Sep 2022

- Accelerated the ACE-T (Agent Commission Engine – Tiering) by refactoring Oracle PL/SQL, indexing high-traffic tables, and resequencing Control-M jobs cut runtime **60% and eliminated $200K/year** in overpayments for 3,000 agents.
- Implemented **OAuth2-based JWT** authentication with granular role-based access control (**RBAC**) in Spring Boot services, driving a **95%** reduction in unauthorized-access incidents and enabling a HIPAA audit(0 findings).
- Designed and shipped an **Angular + Spring MVC** dashboard that streams payout and tiering metrics via WebSocket push **cut average load time 5s to 1.2s** and **boosted engagement by 18%** across **65,000** staff.

Smart Stream Technologies, India | Software Engineer Intern Dec 2020 - Jun 2021

- Developed a Java 17 GraphQL API with Spring Boot and a React SPA for SmartStream's cash-reconciliation portal, applying **GoF design patterns** and **TDD** to minimized client network calls by 70% and lift daily active sessions by 45%.
- Achieved **99.9% uptime** for **25K+ real-time transit updates/min** by automating blue-green deployments on Azure Kubernetes Service with Istio traffic shifting and Terraform-managed infrastructure.

Projects

CollabDocs | System Design, React, Yjs (CRDT), Firebase, WebSockets, AWS ECS, Docker

- Architected an **open-source real-time** CRDT editor with offline sync, whose Dockerized WebSocket backend on AWS ECS cut **latency by 300ms** and Firestore + Yjs merging trimmed memory use **42%**, enabling seamless cross-device collaboration.

Scalable Payment Gateway | Spring Boot, Kafka, PostgreSQL, Redis, Docker, Prometheus

- Built a **fault-tolerant** payment gateway of Dockerized Spring Boot microservices with real-time Kafka-stream **fraud detection and Prometheus observability**, boosting fraud-flag accuracy to **98%** and reducing incident-response time **60%**.

MentorBridge | React.js, TypeScript, Material-UI, Node.js, Express.js, MySQL

- Developed a **full-stack** mentorship platform that unifies course selection, mentor matching, and progress dashboards, delivering a streamlined component-based interface that improved user retention by 72%.

Technical Skills

Languages: Java, Python, JavaScript, TypeScript, PHP, HTML, CSS, Shell Scripting
Databases: PostgreSQL, Microsoft SQL Server, MySQL, MongoDB, DynamoDB, Redis
Cloud & DevOps: AWS (EC2, S3, Lambda, SQS, ElastiCache), Docker, Kubernetes, Jenkins, Git
Frameworks & Libraries: Spring Boot, Hibernate, Node.js, React, Angular, Ajax, Maven, RESTful APIs, Pandas
Testing & QA: JUnit, Mockito, Selenium, Postman, REST Assured, SonarQube
Tools & Platforms: Linux, Apache Kafka, Bitbucket, Splunk, ServiceNow, SoapUI, Figma, Jira, Agile/Scrum
Certification: AWS Certified Cloud Practitioner

Achievements

- Developed data-driven HRT/breast cancer analysis pipeline with Cosmos data, securing 2nd place at **Epic Cosmos Hackathon**.
- **Elected GSA Member**, UIC (2024–2025), Reviewed policies, coordinated workshops, allocated student funding.
- **Led** "ZIGBEE" IoT Club, expanding membership to over **150+** students and facilitating hands-on Zigbee workshops and hackathons.
- Published and presented **IEEE paper** on IOT-based Accident Reporting System system at INCET 2021 conference.